



REALTECH STP

Simple MQTT and SFTP Guide

Missing trip notification only

Only the physical RFID-reader trip is uploaded and counted

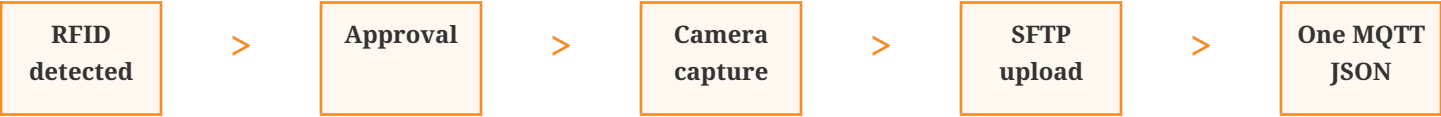
Version	Application v0.1.43
Main change	No synthetic missing trip
Topic	IWS_STP-001/Device_Response
!	Main understanding A missing IN or OUT is now only information inside the current physical trip JSON. It is not a separate trip, image, pending item or processed item.

1. What changed in v0.1.43

Simple comparison of the old and new missing-trip behavior

Old behavior	New behavior
A synthetic missing trip was created.	No synthetic trip is created.
Two trip rows could be pending/processed.	Only one physical trip is pending/processed.
The missing trip could be published separately.	Missing direction is notified inside the physical trip JSON.
Synthetic trip had no image.	The physical trip uses normal camera and SFTP upload.

Physical trip rule



- IN reader detection means direction is IN.
- OUT reader detection means direction is OUT.
- If the opposite trip is missing, the current physical trip still follows the normal image and upload process.
- The missing opposite direction is included as a notification in the same MQTT JSON.

No second upload

There is no second image, no second SFTP upload and no second MQTT transaction for the missing direction.

2. Pending and Processed counting

Dashboard counts only real RFID-reader transactions

Situation	Pending count	Processed count
Normal physical IN or OUT	+1 before sync	+1 after successful MQTT publish
Exceptional physical IN or OUT	+1 before sync	+1 after successful MQTT publish
Missing IN/OUT notification	No change	No change
Legacy synthetic reconciliation row	Excluded	Excluded

Example

An OUT reader detects a vehicle, but no earlier IN exists. After approval, only the OUT trip is saved. The dashboard may show Pending +1, then Processed +1 after synchronization. The missing IN notification does not add another count.

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Legacy database cleanup

At startup, older NotCaptured-ExceptionalReconciliation rows are archived internally. They are not synchronized and are removed from Pending and Processed dashboard totals.

Internal pairing rule

- Missing OUT: the older active IN is closed internally. No OUT transaction is inserted.
- Missing IN: the current OUT is stored without a synthetic IN and without a fake entryTransactionId.

3. Missing IN - one OUT MQTT message

OUT reader detects the RFID, but no active IN trip exists

After Manual or Auto approval, the OUT image is captured and uploaded normally. The same OUT JSON contains both the missing-trip notification and approval details.

```
{
  "topic": "IWS_STP-001/Device_Response",
  "qos": 1,
  "payload": {
    "schemaVersion": "1.1",
    "transactionId": "d58c3243-adab-4f20-b435-6a33e9579231",
    "siteId": "Manurankulam STP Yard",
    "deviceId": "IWS STP 001",
    "laneId": "Manurankulam Tambaram",
    "deviceName": "STP-IWS-001",
    "direction": "OUT",
    "rfidNumber": "E280691500004020ED9648B3",
    "vehicleNumber": "TN 56 RTS 002",
    "accessType": "Free",
    "previousBalance": 0,
    "debitAmount": 0,
    "newBalance": 0,
    "imageCaptureStatus": "Captured",
    "imageFileName": "Manurankulam_STP_Yard - 2026-07-27_14-07-11 - OUT.png",
    "imageRemotePath": "/home/ftpuser/ftp/STP/Tambaram/STP-IWS-001/Manurankulam_STP_Yard - 2026-07-27_14-07-11 - OUT.png",
    "tripDate": "2026-07-27",
    "tripTime": "14:07:11.627",
    "timeZoneOffset": "+05:30",
    "processedAt": "2026-07-27T14:07:11.6277517+05:30",
    "missingTripNotification": {
      "missingDirection": "IN",
      "notificationOnly": true,
      "syntheticTripCreated": false,
      "countedAsTransaction": false,
      "message": "Expected IN trip was missing. Notification only; no separate missing trip was created, uploaded, or counted."
    },
    "exceptionalApproval": {
      "approvalMode": "Manual",
      "reason": "Missing IN Trip",
      "approverName": "Krishna",
      "approverRole": "Engineer",
      "approverMobile": "9944209506",
      "approvalDate": "2026-07-27",
      "approvalTime": "14:06:55.892",
      "timeZoneOffset": "+05:30"
    },
    "notes": "Current OUT trip approved. Missing IN was reported to the server; no synthetic IN trip was created."
  }
}
```



Important

entryTransactionId is omitted because there is no real IN trip. Only this OUT is counted.

4. Missing OUT - one IN MQTT message

IN reader detects the RFID while an earlier IN is still active

After approval, the earlier IN is closed internally. The new physical IN image is captured and uploaded. No synthetic OUT transaction is created.

```
{
  "topic": "IWS_STP-001/Device_Response",
  "qos": 1,
  "payload": {
    "schemaVersion": "1.1",
    "transactionId": "46bc7046-530c-4242-9913-87b2d19f61d0",
    "siteId": "Manurankulam STP Yard",
    "deviceId": "IWS STP 001",
    "laneId": "Manurankulam Tambaram",
    "deviceName": "STP-IWS-001",
    "direction": "IN",
    "rfidNumber": "E2801191A5040076A582E96B",
    "vehicleNumber": "TN 56 R 1899",
    "accessType": "Paid",
    "previousBalance": 10000,
    "debitAmount": 100,
    "newBalance": 9900,
    "imageCaptureStatus": "Captured",
    "imageFileName": "Manurankulam_STP_Yard - 2026-07-27_10-59-57 - IN.png",
    "imageRemotePath": "/home/ftpuser/ftp/STP/Tambaram/STP-IWS-001/Manurankulam_STP_Yard - 2026-07-27_10-59-57 - IN.png",
    "tripDate": "2026-07-27",
    "tripTime": "10:59:57.918",
    "timeZoneOffset": "+05:30",
    "processedAt": "2026-07-27T10:59:57.9187521+05:30",
    "missingTripNotification": {
      "missingDirection": "OUT",
      "notificationOnly": true,
      "syntheticTripCreated": false,
      "countedAsTransaction": false,
      "message": "Expected OUT trip was missing. Notification only; no separate missing trip was created, uploaded, or counted."
    },
    "exceptionalApproval": {
      "approvalMode": "Auto",
      "reason": "Missing OUT Trip",
      "approverName": "SYSTEM",
      "approverRole": "IN AUTO APPROVAL",
      "approverMobile": "N/A",
      "approvalDate": "2026-07-27",
      "approvalTime": "10:59:46.534",
      "timeZoneOffset": "+05:30"
    },
    "notes": "Current IN trip approved. Missing OUT was reported to the server; no synthetic OUT trip was created."
  }
}
```

Count result

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Only the new physical IN becomes Pending and later Processed. The missing OUT changes neither count.

5. Normal IN and OUT behavior

No missing-trip fields are sent when the sequence is complete

Normal IN

```
{
  "topic": "IWS_STP-001/Device_Response",
  "qos": 1,
  "payload": {
    "schemaVersion": "1.1",
    "transactionId": "2663656e-1fa6-4696-b93c-3f46662572aa",
    "siteId": "Manurankulam STP Yard",
    "deviceId": "IWS STP 001",
    "laneId": "Manurankulam Tambaram",
    "deviceName": "STP-IWS-001",
    "direction": "IN",
    "rfidNumber": "E2801191A5040076A582E96B",
    "vehicleNumber": "TN 56 R 1899",
    "accessType": "Paid",
    "previousBalance": 9800,
    "debitAmount": 100,
    "newBalance": 9700,
    "imageCaptureStatus": "Captured",
    "imageFileName": "...IN.png",
    "imageRemotePath": "/home/.../...IN.png"
  }
}
```

Normal OUT

```
{
  "topic": "IWS_STP-001/Device_Response",
  "qos": 1,
  "payload": {
    "schemaVersion": "1.1",
    "transactionId": "9a3b5b9e-9c87-420d-abcb-bd2f5f7b8954",
    "siteId": "Manurankulam STP Yard",
    "deviceId": "IWS STP 001",
    "laneId": "Manurankulam Tambaram",
    "deviceName": "STP-IWS-001",
    "direction": "OUT",
    "entryTransactionId": "2663656e-1fa6-4696-b93c-3f46662572aa",
    "rfidNumber": "E2801191A5040076A582E96B",
    "vehicleNumber": "TN 56 R 1899",
    "accessType": "Paid",
    "previousBalance": 9700,
    "debitAmount": 0,
    "newBalance": 9700,
    "imageCaptureStatus": "Captured",
    "imageFileName": "...OUT.png",
    "imageRemotePath": "/home/.../...OUT.png"
  }
}
```

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Normal sequence

The normal OUT contains the real IN transaction ID. missingTripNotification and exceptionalApproval are omitted.

6. Recharge messages

Recharge behavior is unchanged

Server sends recharge request

```
{
  "topic": "IWS_STP-001/STP_RECHARGE_FROM_SERVER",
  "qos": 1,
  "payload": {
    "schemaVersion": "1.0",
    "messageType": "rfidRecharge",
    "rechargeId": "RCH-20260727-000001",
    "siteId": "Manurankulam STP Yard",
    "deviceId": "IWS STP 001",
    "rfidNumber": "E2801191A5040076A582E96B",
    "vehicleNumber": "TN 56 R 1899",
    "previousBalance": 9700,
    "rechargeAmount": 300,
    "newBalance": 10000,
    "paymentReference": "PAY-20260727-77881"
  }
}
```

Device confirms Applied

```
{
  "topic": "IWS_STP-001/STP_RECHARGE_ACK_FROM_DEVICE",
  "qos": 1,
  "payload": {
    "schemaVersion": "1.0",
    "messageType": "rfidRechargeAck",
    "rechargeId": "RCH-20260727-000001",
    "siteId": "Manurankulam STP Yard",
    "deviceId": "IWS STP 001",
    "laneId": "Manurankulam Tambaram",
    "deviceName": "STP-IWS-001",
    "rfidNumber": "E2801191A5040076A582E96B",
    "vehicleNumber": "TN 56 R 1899",
    "status": "Applied",
    "success": true,
    "duplicate": false,
    "rechargeAmount": 300,
    "previousBalance": 9700,
    "newBalance": 10000,
    "serverBalanceMatched": true,
    "message": "Recharge applied to the local RFID balance."
  }
}
```

7. SFTP image upload

Only the current physical trip image is uploaded



For both normal and approved exceptional trips, the actual IN/OUT camera image follows the same SFTP process.

Case	Image upload
Normal IN or OUT	Upload the physical trip image.
Missing IN approved at OUT reader	Upload only the OUT image.
Missing OUT approved at IN reader	Upload only the new IN image.
Missing direction notification	No separate image or upload.

```
{
  "protocol": "SFTP",
  "host": "gcam-ftp.rtsiot.com",
  "port": 22,
  "remoteDirectory": "/home/ftpuser/ftp/STP/Tambaram/STP-IWS-001",
  "fileName": "Manurankulam_STP_Yard - 2026-07-27_14-07-11 - OUT.png",
  "overwrite": true
}
```

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Upload order
The current physical image is uploaded first. Then one MQTT JSON is sent with imageRemotePath and any missing-trip notification.

8. Application screen behavior

Server status, pending clear and startup loading remain available

Feature	Plain meaning
INTERNET status	Shows whether general internet access is available.
SERVER status	Shows MQTT/SFTP connectivity beside Internet Status.
PROCESSED	Counts successfully synchronized physical trips only.
PENDING	Counts unsynchronized physical trips only.
Data > Clear Pending Data	Deletes the visible pending physical trips after confirmation.
Startup loading window	Shows configuration, storage, database, services and dashboard loading progress.



Missing notifications are invisible to the counters

A missing IN or OUT notification is metadata inside the current trip. It never appears as an extra Pending or Processed number.

9. Simple server checklist

Use this as the final integration understanding

No.	Check
1	Subscribe to IWS_STP-001/Device_Response.
2	Treat transactionId as the unique ID of the physical trip.
3	Read direction from the reader-generated IN or OUT value.
4	When missingTripNotification is present, record the missing direction as information only.
5	Do not create a second server trip from the missing notification.
6	Do not expect a second image or SFTP file for the missing direction.
7	Use exceptionalApproval for Manual/Auto mode, name, role, mobile and approval time.
8	For missing IN, entryTransactionId is omitted.
9	For missing OUT, the previous IN is closed internally; only the new IN is published.
10	Count only the physical MQTT transaction once.

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Final understanding

One RFID-reader event = one physical trip = one image = one MQTT transaction = one Pending/Processed count. A missing opposite trip is notification data only.